

1317. *Proposed by Ángel Plaza, Universidad de Las Palmas de Gran Canaria, Spain.*

Calculate the following sum, with proof:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \left(\frac{1}{2n-1} - \frac{1}{2n} + \frac{1}{2n+1} - \cdots \right).$$

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Jayden Brown-Akoto (student author)

The desired sum is shown to be $\ln(2) - \frac{3 \cdot (\ln(2))^2}{4} - \frac{\pi}{2} + \frac{\pi^2}{16}$.

Let

$$\begin{aligned} I &:= \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \left(\frac{1}{2n-1} - \frac{1}{2n} + \frac{1}{2n+1} - \cdots \right) \\ &= \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n(2n-1)} - \frac{(-1)^n}{n(2n)} + \frac{(-1)^n}{n(2n+1)} - \cdots \right). \end{aligned}$$

Additionally, let

$$\begin{aligned} f(x) &:= \sum_{n=1}^{\infty} \left(\frac{(-1)^n x^{2n-1}}{n(2n-1)} - \frac{(-1)^n x^{2n}}{n(2n)} + \frac{(-1)^n x^{2n+1}}{n(2n+1)} - \cdots \right) \\ &= \sum_{n=1}^{\infty} (-1)^n x^{2n-2} \left(\frac{x^1}{n(2n-1)} - \frac{x^2}{n(2n)} + \frac{x^3}{n(2n+1)} - \cdots \right). \end{aligned}$$

We note that $f(1) = \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n(2n-1)} - \frac{(-1)^n}{n(2n)} + \frac{(-1)^n}{n(2n+1)} - \cdots \right) = I$ and $f(0) = \sum_{n=1}^{\infty} \left(\frac{0}{n(2n-1)} - \frac{0}{n(2n)} + \frac{0}{n(2n+1)} - \cdots \right) = \sum_{n=1}^{\infty} 0 = 0$.

Finally, we introduce the parameter k and let

$$\begin{aligned} u(x) &:= \sum_{n=1}^{\infty} \left(\frac{(-1)^n x^{2n-2}}{n} - \frac{(-1)^n x^{2n-1}}{n} + \frac{(-1)^n x^{2n}}{n} - \cdots \right) \\ &= \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \left(\frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \right). \end{aligned}$$

Lemma 1: $f(x)$ is uniformly convergent on $[0, 1]$.

To prove uniform convergence of $f(x)$ over $[0, 1]$, we will show that $f(x)$ satisfies the Weierstrass M-Test.

To begin, let

$$\begin{aligned} s_n(x) &:= \left(\frac{x^1}{n(2n-1)} - \frac{x^2}{n(2n)} + \frac{x^3}{n(2n+1)} - \dots \right) \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^k}{n(2n+k-2)} \end{aligned}$$

for all $n \geq 1$. From the above definition, it is clear that each $s_n(x)$ can be represented as a power series $\sum_{k=1}^{\infty} a_k x^k$ with $a_k = \frac{(-1)^{k+1}}{n(2n+k-2)}$. Thus, it can be concluded that each $s_n(x)$ is uniformly convergent on its convergence interval. In addition, each $s_n(x)$ is an alternating series. So, the respective convergence interval for any $s_n(x)$ is also the interval on which it satisfies the alternating series test. Now,

$$\lim_{k \rightarrow \infty} \left| \frac{(-1)^{k+1} \cdot x^k}{n(2n+k-2)} \right| = \lim_{k \rightarrow \infty} \left| \frac{x^k}{n(2n+k-2)} \right|.$$

From this, it is clear that $\lim_{k \rightarrow \infty} \left| \frac{(-1)^{k+1} \cdot x^k}{n(2n+k-2)} \right| = 0$ when $-1 \leq x \leq 1$. Additionally, $\left| \frac{(-1)^{k+1} \cdot x^k}{n(2n+k-2)} \right| > \left| \frac{(-1)^{k+2} \cdot x^{k+1}}{n(2n+k-1)} \right|$ for all $k \geq 1$ $[-1, 0)$ and $(0, 1]$. Moreover, at $x = 0$, $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^k}{n(2n+k-2)} = 0$. Thus, each $s_n(x)$ is uniformly convergent on $[-1, 1]$.

As proved earlier, when $-1 \leq x \leq 1$, the sequence $\left\{ \left| \frac{(-1)^{k+1} \cdot x^{2n+k-2}}{n(2n+k-2)} \right| \right\}$ for $n \in \mathbb{N}^*$ decreases monotonically or is 0 for all $n \geq 1$. So if we let $S_{2n+1}(x)$ denote the $(2n+1)th$ partial sum of a specific $s_n(x)$ and $S_{2n}(x)$ denote the $(2n)th$, we get that on $[0, 1]$,

$$\begin{aligned} \lim_{n \rightarrow \infty} S_{2n+1}(x) &= \frac{x^{2n-1}}{n(2n-1)} - \left(\frac{x^{2n}}{n(2n)} - \frac{x^{2n+1}}{n(2n+1)} \right) - \dots \leq \frac{x^{2n-1}}{n(2n-1)} \text{ and} \\ \lim_{n \rightarrow \infty} S_{2n+1}(x) &= \left(\frac{x^{2n-1}}{n(2n-1)} - \frac{x^{2n}}{n(2n)} \right) + \left(\frac{x^{2n+1}}{n(2n+1)} - \frac{x^{2n+2}}{n(2n+2)} \right) + \dots \geq 0. \end{aligned}$$

And, since each $s_n(x)$ converges, the two limits are equal to $s_n(x)$. Thus,

$$0 \leq s_n(x) \leq \frac{x^{2n-1}}{n(2n-1)}.$$

As shown earlier, $0 \leq s_n(x) \leq \frac{x}{n(2n-1)}$ on $[0, 1]$ for each $n \geq 1$. Additionally, since x^{2n-2} is nonnegative on the interval $[0, 1]$, $0 \leq s_n(x) \cdot x^{2n-2} \leq \frac{x^{2n-1}}{n(2n-1)}$

on $[0, 1]$. Now, $\lim_{n \rightarrow \infty} \frac{x^{2n-1}}{n(2n-1)} = 0$ when $0 \leq x \leq 1$. So, as $n \rightarrow \infty$, $s_n(x) \cdot x^{2n-1} \rightarrow 0$ on $0 \leq x \leq 1$ on the same interval. Furthermore, since $s_n(x) = \left(\frac{x^1}{n(2n-1)} - \frac{x^2}{n(2n)} + \frac{x^3}{n(2n+1)} - \dots \right)$ and is uniformly convergent on $[0, 1]$, $f(x) = \sum_{n=1}^{\infty} (-1)^n \cdot s_n(x) \cdot x^{2n-2}$ on the interval $[0, 1]$.

Since $s_n(x) \cdot x^{2n-2} \geq 0$ on $[0, 1]$, $|(-1)^n s_n(x) \cdot x^{2n-2}| = s_n(x) \cdot x^{2n-2}$ on $[0, 1]$. Additionally, $s_n(x) \cdot x^{2n-2} \leq \frac{x^{2n-1}}{n(2n-1)} \leq \frac{1}{n(2n-1)} \leq \frac{1}{n^2}$ on $[0, 1]$ for all $n \geq 1$. Thus $|(-1)^n s_n(x) \cdot x^{2n-2}| \leq \frac{1}{n^2}$ for all $n \geq 1$ on $[0, 1]$. Moreover, since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, $f(x) = \sum_{n=1}^{\infty} (-1)^n \cdot s_n(x) \cdot x^{2n-2}$ uniformly converges on $[0, 1]$ by the Weierstrass M-Test.

Lemma 2: $u(x)$ is uniformly convergent and $u(x) = f'(x)$ on $[0, 1]$.

To prove uniform convergence of $u(x)$ over $[0, 1]$, we will show that $u(x)$ satisfies the definition for uniform convergence. The equality of $u(x)$ and $f'(x)$ over $[0, 1]$ follows from the uniform convergence of $u(x)$ over $[0, 1)$ and the earlier proven uniform convergence of $f(x)$ over $[0, 1]$.

Term-by-term differentiation of $f(x)$ produces the following sum:

$$\sum_{n=1}^{\infty} \frac{d}{dx} ((-1)^n \cdot s_n(x) \cdot x^{2n-2}) = \sum_{n=1}^{\infty} (-1)^n (s'_n(x) \cdot x^{2n-2} + s_n(x) \cdot (2n-2)x^{2n-3}).$$

Since $s_n(x)$ is a power series, $s'_n(x)$ is equal to the term-by-term differentiation of $s_n(x)$ and has the same radius of convergence as $s_n(x)$. Thus,

$$\begin{aligned} s'_n(x) &= \left(\frac{1}{n(2n-1)} - \frac{2x}{n(2n)} + \frac{3x^2}{n(2n+1)} - \dots \right) \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot k \cdot x^{k-1}}{n(2n+k-2)}. \end{aligned}$$

Substituting $s_n(x)$ and $s'_n(x)$ into the term-by-term differentiation of $f(x)$, $s'_n(x) \cdot x^{2n-2} + s_n(x) \cdot (2n-2)x^{2n-3}$ becomes:

$$\begin{aligned}
& \left(\sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot k \cdot x^{k-1}}{n(2n+k-2)} \right) \cdot x^{2n-2} + \left(\sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^k}{n(2n+k-2)} \right) \cdot (2n-2)x^{2n-3} \\
&= \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot k \cdot x^{2n+k-3}}{n(2n+k-2)} + \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot (2n-2) \cdot x^{2n+k-3}}{n(2n+k-2)} \\
&= \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot k \cdot x^{2n+k-3} + (-1)^{k+1} \cdot (2n-2) \cdot x^{2n+k-3}}{n(2n+k-2)} \\
&= \sum_{k=1}^{\infty} \frac{(2n+k-2)(-1)^{k+1} \cdot x^{2n+k-3}}{n(2n+k-2)} \\
&= \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^{2n+k-3}}{n}.
\end{aligned}$$

Thus,

$$\begin{aligned}
& \sum_{n=1}^{\infty} \frac{d}{dx} \left((-1)^n \cdot s_n(x) \cdot x^{2n-2} \right) \\
&= \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \\
&= u(x).
\end{aligned}$$

Now, $u(x)$ can be arranged such that:

$$u(x) = \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-2}}{n} (1 - x + x^2 - \cdots).$$

And on $[0, 1)$,

$$u(x) = \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-2}}{n(1+x)}.$$

Now, on $[0, 1)$, x^{2n-2} decreases with increasing n or is 0 for $n \geq 1$. Thus, $\left\{ \left| \frac{(-1)^n x^{2n-2}}{n(1+x)} \right| \right\}$ decreases monotonically or is 0 as $n \rightarrow \infty$. By the same reasoning, $\lim_{n \rightarrow \infty} \left| \frac{(-1)^n x^{2n-2}}{n(1+x)} \right| = 0$. Since $\left\{ \left| \frac{(-1)^n x^{2n-2}}{n(1+x)} \right| \right\}$ monotonically decreases to 0, $u(x)$ is a convergent alternating series and the alternating series remainder theorem can be applied. That is, if we let $U_N(x)$ denote $\sum_{n=1}^N \frac{(-1)^n x^{2n-2}}{n(1+x)}$, $|U_N(x) - u(x)| \leq \frac{x^{2N}}{(N+1)(1+x)}$.

Now, let $n > N$. Then, on $[0, 1]$, $|U_n(x) - u(x)| \leq \frac{x^{2n}}{(n+1)(x+1)} \leq \frac{x^{2N}}{(N+1)(x+1)}$, by the alternating series remainder theorem and the monotone decreasing property of $\left\{ \left| \frac{(-1)^n x^{2n-2}}{n(1+x)} \right| \right\}$. Furthermore, on $[0, 1]$, $\frac{x^{2N}}{(N+1)(x+1)} < \frac{1}{2(N+1)} < \frac{1}{N}$. So, on $[0, 1]$, it can be said that for $t > 0$, if $N > \frac{1}{t}$, then $|U_n(x) - u(x)| < t$ for all $n > N$. Thus, $u(x)$ is uniformly convergent on $[0, 1]$ by definition of uniform convergence. Moreover, since $f(x)$ is uniformly convergent on $[0, 1]$, $u(x) = f'(x)$ on $[0, 1]$. From now on, $u(x)$ will be denoted as $f'(x)$.

Solution:

To begin, we note

$$\begin{aligned} & \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \int_0^1 \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} dx \\ &= \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{(-1)^{n+k+1}}{n(2n+k-2)} \\ &= \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n(2n-1)} - \frac{(-1)^n}{n(2n)} + \frac{(-1)^n}{n(2n+1)} - \dots \right) \\ &= I. \end{aligned}$$

Now, we let $S_K(x) := \sum_{k=1}^K \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n}$ with $0 \leq x < 1$. We observe that each $S_K(x)$ is measurable for all $n \geq 1$, since a finite sum of measurable functions is also measurable. Additionally, we observe that the terms of $\left\{ \left| \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \right| \right\}$ decrease monotonically to 0 as $k \rightarrow \infty$ for all $n \geq 1$ on the interval $(0, 1)$. We add that at $x = 0$, the terms of $\left\{ \left| \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \right| \right\}$ are all 0. From this we can conclude that for odd terms $|S_{2K+1}(x)|$, $1 \geq |S_{2K+1}(x)|$ since $1 \geq \left| \frac{(-1)^n \cdot x^{2n-2}}{n} \right| \geq \left| \frac{(-1)^n \cdot x^{2n-2}}{n} - \dots - \left(\frac{(-1)^n \cdot x^{2n+2K-3}}{n} - \frac{(-1)^n \cdot x^{2n+2K-2}}{n} \right) \right|$. Additionally, we can conclude that for even terms $|S_{2K}(x)|$, $1 \geq |S_{2K}(x)|$ since $1 \geq \left| \frac{(-1)^n \cdot x^{2n-2}}{n} \right| \geq \left| \frac{(-1)^n \cdot x^{2n-2}}{n} - \left(\frac{(-1)^n \cdot x^{2n-1}}{n} - \frac{(-1)^n \cdot x^{2n}}{n} \right) - \dots - \left(\frac{(-1)^n \cdot x^{2n+2K-3}}{n} \right) \right|$. Thus, $|S_K(x)| \leq 1$ for all x on the interval $[0, 1]$ and for all $K \geq 1$. Now, since it has been proven that $|S_K(x)| \leq 1$ for almost all x on $[0, 1]$ ¹ and for all $K \geq 1$, we can apply the Dominating Convergence Theorem (since $g(x) = 1$ is Lebesgue integrable on $[0, 1]$) to conclude that $\sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \int_0^1 \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} dx = \sum_{n=1}^{\infty} \int_0^1 \left(\sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \right) dx$.

¹In fact, $|S_K(x)| \leq 1$ for all x on $[0, 1]$

Similarly, we let $f'_N(x) := \sum_{n=1}^N \sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n}$. Evaluating the inner series under the assumption $|x| < 1$ yields:

$$\begin{aligned} f'_N(x) &= \sum_{n=1}^N \sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \\ &= \sum_{n=1}^N \left(\frac{(-1)^n x^{2n-2}}{n} \sum_{k=1}^{\infty} (-1)^{k+1} \cdot x^{k-1} \right) \\ &= \sum_{n=1}^N \frac{(-1)^n x^{2n-2}}{n(1+x)}. \end{aligned}$$

We can then observe that each $f'_N(x)$ is measurable, since a finite sum of measurable functions is also measurable. Additionally, since each $f'_N(x)$ is an alternating series and the first term of each $f'_N(x) = -\frac{1}{1+x}$. We can apply similar logic as used in the previous case to conclude $|f'_N(x)| \leq 1$ for all x on $[0, 1]$, since $\frac{1}{1+x} \leq 1$ when $|x| < 1$. Now, since $|f'_N(x)| \leq 1$ almost everywhere on $[0, 1]$ and $f'_N(x)$ converges pointwise ² to $f(x)$ almost everywhere on $[0, 1]$, we can apply the the Dominating Convergence Theorem again to conclude that $\sum_{n=1}^{\infty} \int_0^1 \left(\sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \right) dx = \int_0^1 \left(\sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \right) dx$.

Thus applying the Dominating Convergence Theorem twice gives $\int_0^1 \left(\sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \right) dx = \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \int_0^1 \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} dx = I$.

Now, we evaluate $f'(x)$.

²Uniform convergence implies pointwise convergence

Case 1 ($0 < x < 1$)

$$\begin{aligned}
f'(x) &= \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \\
&= \sum_{n=1}^{\infty} \left(\frac{(-1)^n x^{2n-2}}{n} \sum_{k=1}^{\infty} (-1)^{k+1} \cdot x^{k-1} \right) \\
&= \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-2}}{n(1+x)} \\
&= \frac{1}{x^2(x+1)} \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{n} \\
&= -\frac{1}{x^2(x+1)} \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n}}{n} \\
&= -\frac{\ln(x^2+1)}{x^2(x+1)}
\end{aligned}$$

Case 2 ($x = 0$)

$$\begin{aligned}
f'(x) &= \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{(-1)^{n+k+1} \cdot x^{2n+k-3}}{n} \\
&= \sum_{n=1}^{\infty} \left(\frac{(-1)^n x^{2n-2}}{n} \sum_{k=1}^{\infty} (-1)^{k+1} \cdot x^{k-1} \right) \\
&= \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-2}}{n(1+x)} \\
&= \frac{(-1)^1 \cdot 0^0}{1} + \sum_{n=2}^{\infty} \frac{(-1)^n 0^{2n-2}}{n(1+x)} \\
&= -1 + 0 \\
&= -1
\end{aligned}$$

Thus,

$$f'(x) = \begin{cases} -1 & x = 0 \\ -\frac{\ln(x^2+1)}{x^2(x+1)} & 0 < x < 1. \end{cases}$$

We note that $f'(x)$ is continuous on $[0, 1)$.³

³Applying L'Hospital's Rule: $\lim_{x \rightarrow 0} -\frac{\ln(x^2+1)}{x^2(x+1)} = \lim_{x \rightarrow 0} -\frac{2x}{(3x^2+2x)(x^2+1)}$
 $= \lim_{x \rightarrow 0} -\frac{2}{(3x+2)(x^2+1)} = -1 = \lim_{x \rightarrow 0^+} -\frac{\ln(x^2+1)}{x^2(x+1)}$. Additionally, the uniform convergence of $f'(x)$ on $[0, 1)$ implies continuity on $[0, 1)$.

Since removal of a singular point does not affect the integral of a function, it can be said that $I = \int_0^1 f'(x)dx = \lim_{t \rightarrow 0^+} \int_t^1 -\frac{\ln(x^2 + 1)}{x^2(x + 1)}dx$.

Now, to evaluate the integral, t will be restricted to the interval $(0, 1]$ and a parameter α , restricted to $[0, \infty)$, will be introduced into the integral such that

$$h(\alpha) = - \lim_{t \rightarrow 0^+} \int_t^1 \frac{\ln(\alpha x^2 + 1)}{x^2(x + 1)}dx.$$

We note that $h(1) = I$ and $h(0) = 0$. Additionally, we observe that the integral is uniformly convergent with respect to α ⁴. As such, $\frac{d}{d\alpha} \int = \int \frac{\partial}{\partial \alpha}$. So, the limit notation will be temporarily omitted for clarity.

Taking the derivative, we get

$$h'(\alpha) = -\frac{d}{d\alpha} \int_t^1 \frac{\ln(\alpha x^2 + 1)}{x^2(x + 1)}dx.$$

Thus,

$$\begin{aligned} h'(\alpha) &= - \int_t^1 \frac{\partial}{\partial \alpha} \frac{\ln(\alpha x^2 + 1)}{x^2(x + 1)}dx \\ &= - \int_t^1 \frac{1}{x^2(x + 1)} \cdot \frac{x^2}{\alpha x^2 + 1}dx \\ &= - \int_t^1 \frac{1}{(\alpha x^2 + 1)(x + 1)}dx \\ &= - \int_t^1 \left(\left(-\frac{\alpha}{1 + \alpha} \right) \left(\frac{x - 1}{\alpha x^2 + 1} \right) + \frac{1}{(\alpha + 1)(x + 1)} \right) dx \\ &= \int_t^1 \left(\frac{\alpha}{1 + \alpha} \right) \left(\frac{x - 1}{\alpha x^2 + 1} \right) dx + - \int_t^1 \frac{1}{(\alpha + 1)(x + 1)}dx \\ &= \left(\frac{\alpha}{1 + \alpha} \right) \left(\int_t^1 \left(\frac{x}{\alpha x^2 + 1} \right) dx - \int_t^1 \left(\frac{1}{\alpha x^2 + 1} \right) dx \right) \\ &\quad + - \frac{1}{\alpha + 1} \int_t^1 \frac{1}{x + 1}dx \\ &= \frac{\alpha}{1 + \alpha} \int_t^1 \frac{x}{\alpha x^2 + 1}dx - \frac{\alpha}{1 + \alpha} \int_t^1 \frac{1}{\alpha x^2 + 1}dx + \\ &\quad - \frac{1}{\alpha + 1} \int_t^1 \frac{1}{x + 1}dx \end{aligned}$$

⁴We know that for every $\epsilon > 0$, there exists r such that $\left| \int_r^1 -\frac{\ln(\alpha x^2 + 1)}{x^2(x + 1)}dx \right| < \epsilon$ for all $\alpha > 0$ since each of these functions is integrable on the interval $[r, 1]$ with $0 < r < 1$. Thus, r can be as arbitrarily close to 1 as needed for $\left| \int_r^1 -\frac{\ln(\alpha x^2 + 1)}{x^2(x + 1)}dx \right| < \epsilon$.

Now, we evaluate each integral separately:

(1)

To evaluate (1) we rearrange the terms and let $u = \alpha x^2 + 1$, so $du = 2\alpha x dx$:

$$\begin{aligned} \frac{\alpha}{1+\alpha} \int_t^1 \frac{x}{\alpha x^2 + 1} dx &= \frac{1}{2(1+\alpha)} \int_t^1 \frac{2\alpha x}{\alpha x^2 + 1} dx = \frac{1}{2(1+\alpha)} \int_{t^2\alpha+1}^{\alpha+1} \frac{du}{u} \\ &= \lim_{t \rightarrow 0^+} \left. \frac{\ln |\alpha x^2 + 1|}{2(1+\alpha)} \right|_t^1 = \frac{\ln |\alpha + 1|}{2(1+\alpha)} = \frac{\ln(\alpha + 1)}{2(\alpha + 1)} \end{aligned}$$

We observe that $\frac{\ln |\alpha + 1|}{2(1+\alpha)} = \frac{\ln(\alpha + 1)}{2(\alpha + 1)}$, since $\alpha \geq 0$.

(2)

$$\begin{aligned} -\frac{\alpha}{1+\alpha} \int_t^1 \frac{1}{\alpha x^2 + 1} dx &= -\frac{1}{1+\alpha} \int_t^1 \frac{1}{x^2 + \frac{1}{\alpha}} dx = -\lim_{t \rightarrow 0^+} \left. \frac{\sqrt{\alpha} \arctan(x\sqrt{\alpha})}{1+\alpha} \right|_t^1 \\ &= -\frac{\sqrt{\alpha} \arctan \sqrt{\alpha}}{1+\alpha} \end{aligned}$$

(3)

$$-\frac{1}{\alpha+1} \int_t^1 \frac{1}{x+1} dx = -\lim_{t \rightarrow 0^+} \left. \frac{\ln |x+1|}{\alpha+1} \right|_t^1 = -\frac{\ln(2)}{\alpha+1}.$$

Recombining the evaluated integrals, we get

$$\begin{aligned} h'(\alpha) &= -\lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{(\alpha x^2 + 1)(x + 1)} dx \\ &= \frac{\ln(\alpha + 1)}{2(\alpha + 1)} - \frac{\sqrt{\alpha} \arctan \sqrt{\alpha}}{1 + \alpha} - \frac{\ln(2)}{\alpha + 1}. \end{aligned}$$

Now, as noted earlier, $h(1) = I$ and $h(0) = 0$. Thus, $h(1) - h(0) = I$ and $\int_0^1 h'(\alpha) d\alpha = I$ since $h'(\alpha)$ is integrable on $0 \leq \alpha \leq 1$.

Thus,

$$\begin{aligned} I &= \int_0^1 \left(\frac{\ln(\alpha+1)}{2(\alpha+1)} - \frac{\sqrt{\alpha} \arctan \sqrt{\alpha}}{1+\alpha} - \frac{\ln(2)}{\alpha+1} \right) d\alpha \\ &= \int_0^1 \frac{\ln(\alpha+1)}{2(\alpha+1)} d\alpha - \int_0^1 \frac{\sqrt{\alpha} \arctan \sqrt{\alpha}}{1+\alpha} d\alpha - \int_0^1 \frac{\ln(2)}{\alpha+1} d\alpha. \end{aligned}$$

We again evaluate each integral separately:

(1)

$$\int_0^1 \frac{\ln(\alpha+1)}{2(\alpha+1)} d\alpha = \frac{1}{2} \int_0^{\ln(2)} u du = \frac{u^2}{4} \Big|_0^{\ln(2)} = \frac{(\ln(2))^2}{4} - 0 = \frac{(\ln(2))^2}{4}$$

(2)

We let $\alpha = \tan^2(u)$, so that $d\alpha = 2 \tan(u) \sec^2(u) du$ and

$$\begin{aligned} - \int_0^1 \frac{\sqrt{\alpha} \arctan \sqrt{\alpha}}{1+\alpha} d\alpha &= -2 \int_0^{\pi/4} \frac{\sqrt{\tan^2(u)} \arctan \sqrt{\tan^2(u)}}{1+\tan^2(u)} \tan(u) \sec^2(u) du \\ &= -2 \int_0^{\pi/4} \frac{\tan^2(u) \sec^2(u) \arctan(\tan(u))}{1+\tan^2(u)} du \\ &= -2 \int_0^{\pi/4} \frac{u \tan^2(u) \sec^2(u)}{\sec^2(u)} du \\ &= -2 \int_0^{\pi/4} u \tan^2(u) du \\ &= -2 \int_0^{\pi/4} u (\sec^2(u) - 1) du \end{aligned}$$

Now, letting $p = u$ and $dq = \sec^2(u) - 1$, we get $dp = du$ and $q = \tan(u) - u$ and integrate by parts such that $\int pdq = pq - \int qdp$.

$$-2 \int_0^{\pi/4} u (\sec^2(u) - 1) du = -2 \left(u(\tan(u) - u) \Big|_0^{\pi/4} - \int_0^{\pi/4} (\tan(u) - u) du \right)$$

Now,

$$\int_0^{\frac{\pi}{4}} (\tan(u) - u) du = \int_0^{\frac{\pi}{4}} \tan(u) du - \int_0^{\frac{\pi}{4}} u du = \int_0^{\frac{\pi}{4}} \frac{\sin(u)}{\cos(u)} du - \frac{u^2}{2} \Big|_0^{\frac{\pi}{4}}$$

Substituting $v = \cos(u)$, we get $dv = -\sin(u)du$

$$\int_0^{\frac{\pi}{4}} \frac{\sin(u)}{\cos(u)} du - \frac{u^2}{2} \Big|_0^{\frac{\pi}{4}} = - \int_1^{\frac{1}{\sqrt{2}}} \frac{dv}{v} - \frac{u^2}{2} \Big|_0^{\frac{\pi}{4}} = - \ln(|\cos(u)|) \Big|_0^{\frac{\pi}{4}} - \frac{u^2}{2} \Big|_0^{\frac{\pi}{4}}$$

Substituting this value back into the original integral, we get

$$\begin{aligned} & -2 \left(u(\tan(u) - u) \Big|_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} (\tan(u) - u) du \right) \\ &= -2 \left(u(\tan(u) - u) \Big|_0^{\frac{\pi}{4}} - \left(-\ln(|\cos(u)|) \Big|_0^{\frac{\pi}{4}} - \frac{u^2}{2} \Big|_0^{\frac{\pi}{4}} \right) \right) \\ &= -2 \left(\frac{\pi}{4} \left(1 - \frac{\pi}{4} \right) - \left(-\ln\left(\frac{1}{\sqrt{2}}\right) - \frac{\pi^2}{32} \right) \right) \\ &= -2 \left(\frac{\pi}{4} - \frac{\pi^2}{16} - \frac{1}{2} \ln(2) + \frac{\pi^2}{32} \right) \\ &= -2 \left(\frac{\pi}{4} - \frac{1}{2} \ln(2) - \frac{\pi^2}{32} \right) \\ &= -\frac{\pi}{2} + \ln(2) + \frac{\pi^2}{16} \end{aligned}$$

(3)

$$- \int_0^1 \frac{\ln(2)}{\alpha + 1} d\alpha = - \ln(2) \ln(|\alpha + 1|) \Big|_0^1 = -(\ln(2))^2.$$

We add the values of the integrals together to get

$$\begin{aligned}
I &= \int_0^1 \frac{\ln(\alpha+1)}{2(\alpha+1)} d\alpha - \int_0^1 \frac{\sqrt{\alpha} \arctan \sqrt{\alpha}}{1+\alpha} d\alpha - \int_0^1 \frac{\ln(2)}{\alpha+1} d\alpha \\
&= \frac{(\ln(2))^2}{4} - \frac{\pi}{2} + \ln(2) + \frac{\pi^2}{16} - (\ln(2))^2 \\
&= \ln(2) - \frac{3 \cdot (\ln(2))^2}{4} - \frac{\pi}{2} + \frac{\pi^2}{16}.
\end{aligned}$$

Thus, it has been shown:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \left(\frac{1}{2n-1} - \frac{1}{2n} + \frac{1}{2n+1} - \dots \right) = \ln(2) - \frac{3 \cdot (\ln(2))^2}{4} - \frac{\pi}{2} + \frac{\pi^2}{16}.$$